Thermal Stress and Strain

Thermal Stress and Strain

- In a homogenous, isotropic material, change in temperature and change in dimensions are linearly related as;

$$\delta_T = \alpha \Delta T L$$

where

α = ***linear coefficient of thermal expansion*** of the material. It represents quantity per degree C, or per degree F.

ΔT = the algebraic change in temperature of the member

L = the original length of the member

δ_T = the algebraic change in the length of the member (thermal displacements)

- The strain from thermal displacements can be expressed as;

$$\varepsilon_T = \alpha \Delta T$$

- In a statically indeterminate member, thermal displacements will be constrained by the supports, producing thermal stresses, which can be expressed as

$$\sigma_T = E\varepsilon_T = E\alpha \Delta T$$

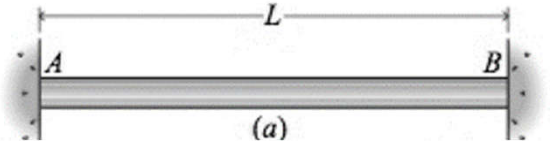
Thermal Stresses and Strains

Example

A 10-m section of a steel pipe [E = 200 GPa and α = ll.9 x 10^{-6})/°C] rail has a cross-sectional area of 7500 mm^2. Both ends of the pipe are in contact with two walls such that, it can be assumed to be rigid, but insignificantly stressed. For an increase in temperature of 50°C, determine

a) The normal stress in the rail.

b) The internal force on a cross section of the rail.



$$\sigma_T = E\varepsilon_T = E\alpha\Delta T$$
$$= (200\times10^{9})(11.9\times10^{-6})(50)$$
$$= 119 \text{ MPa}$$

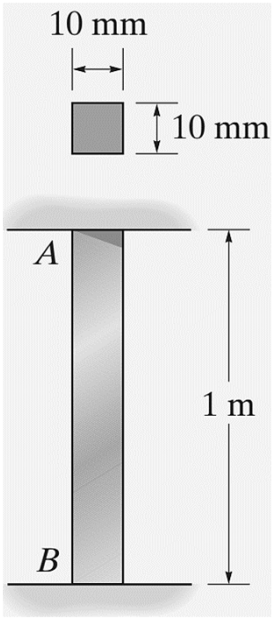
$$\sigma = \frac{F}{A} \Rightarrow F = (119\times10^{6})(7500\times10^{-6})$$
$$\cong 893 \text{ kN}$$

Thermal Stresses and Strains

Example

The A-36 steel bar shown below is constrained to just fit between two fixed supports when $T_1 = 30°C$. If the temperature is raised to $T_2 = 60°C$, determine the average normal thermal stress developed in the bar, and the resulting internal reaction it generates.

Take $\alpha = 12 \times 10^{-6}/°C$ and $E = 200$ GPa.



F = 7.2 kN

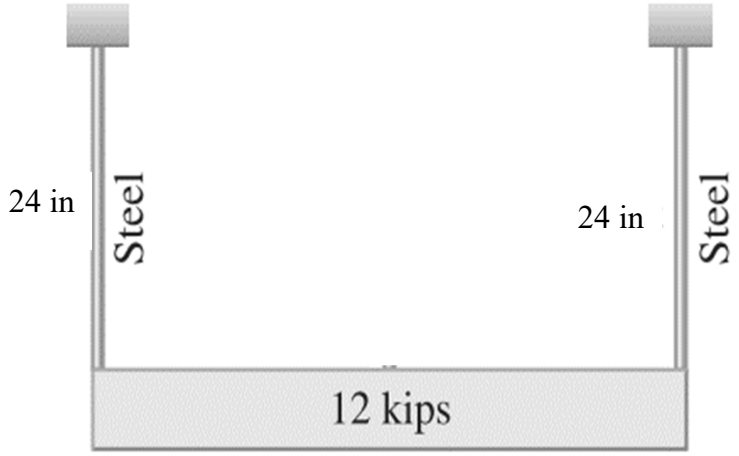
σ = 72 MPa

Thermal Stresses and Strains

Example

A homogeneous, rigid block weighing 12 kips (12000 lb) is suspended off two symmetrically placed rods as shown below. If the weight of the block is distributed equally among the 2 rods, determine the stress in each rod when the temperature of all bars increases by 100°F. Also determine the total change in length of each rod.

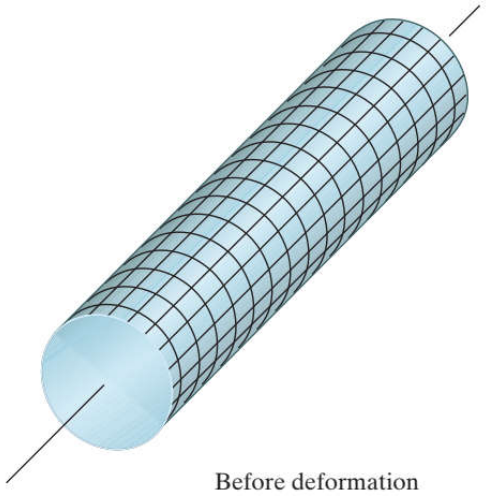
	A (in.^{2})	E (psi)	α (/°F)
Each steel rod	0.75	29×10^6	6.5×10^{-6}



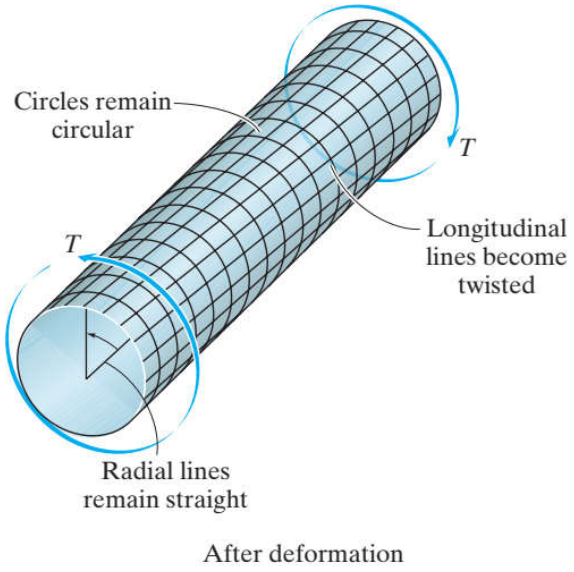
Torsional Stress

Torsional Loading

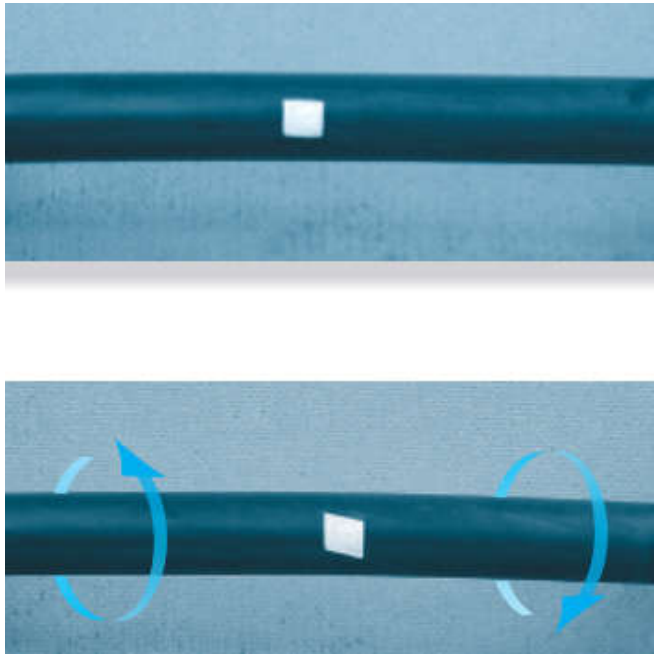
- Torsional loading is from a torque twisting a member about an axis.



Before deformation

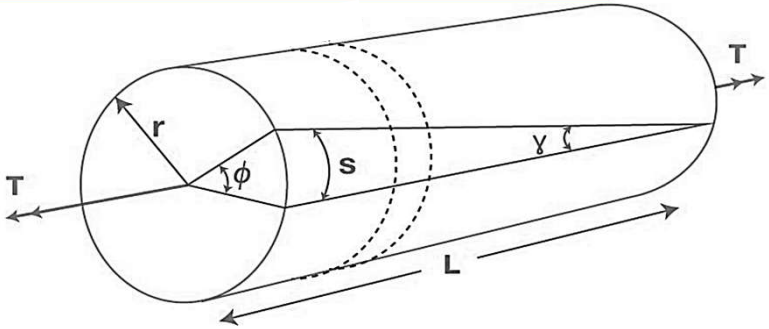


After deformation



Torsional Shear Stress

- Assuming;
  - Pure torsion
  - Circular cross-section that remains plane.



- Within the elastic region, shear stress on the transverse plane due to torsion is calculated as;

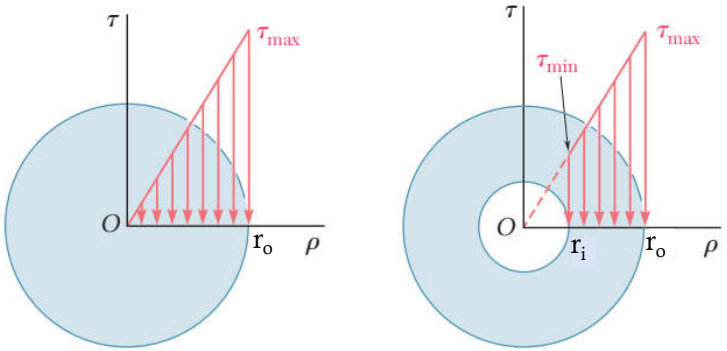
$$\tau_{\max} = \frac{Tr_o}{J}$$

- T is the internal torque..

- J is the polar moment of inertia. For a solid cross-section, it is given as;

$$J = \frac{\pi r^4}{2} = \frac{\pi D^4}{32}$$

- For tubes, stress only occurs on tube thickness



Distribution of shear stress

Power Transmission

- The torque, T (in N.m) on a rotating shaft transmitting an amount of Power P (in Watts) at an angular velocity ω (in rad/s) given by

$$P = T\omega$$

where ω is defined in terms of frequency, f or rate of rotation of the shaft n (in rpm) as

$$\omega = 2\pi f = 2\pi n/60$$

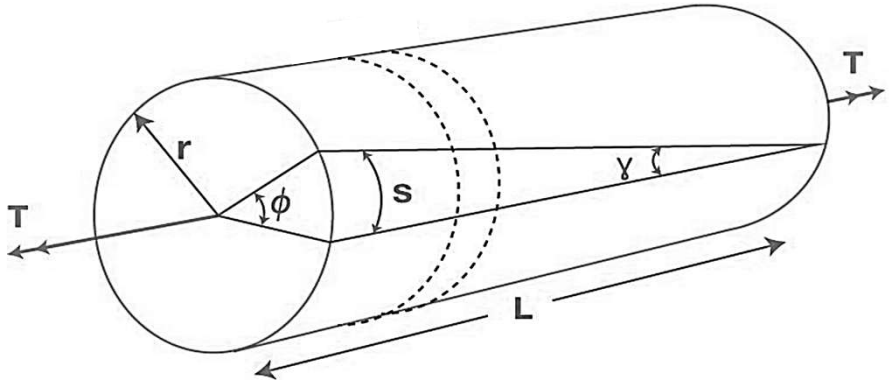
- So when the power to be transmitted and the allowable stress in a shaft are know, an appropriate diameter can be determined for the shaft.

$$\frac{J}{r} = \frac{T}{\tau_{allow}}$$

- Note:
  - 1 horsepower (hp) = 746 Watts
  - 1 hp = 550 ft-lb/s

Torsional Shear Strain

- Assuming;
  - Deformations are through small angles

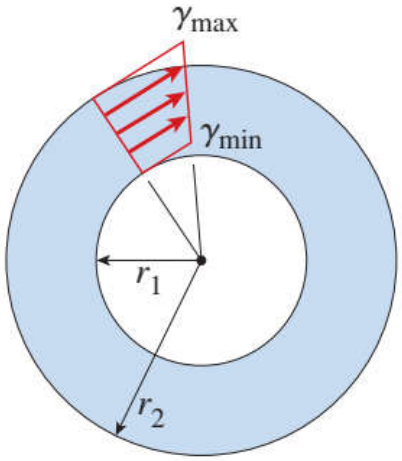


- Shear strain is calculated as

$$\gamma_{max} = \frac{r\phi}{L}$$

- From Hooke's law in shear, deformation is measured as the angle of twist;

$$\phi = \frac{TL}{GJ}$$



- Principle of superposition applies in determining the total deformation in a shaft if required conditions are satisfied.

Torsional Loading: Internal Forces

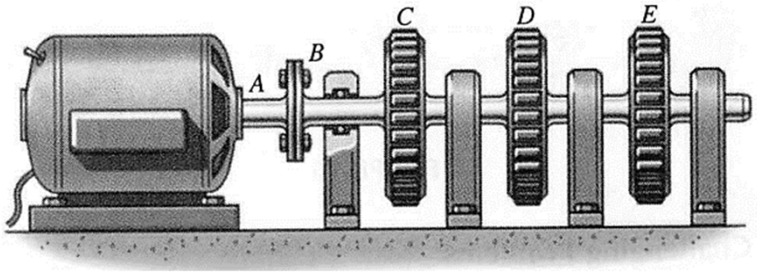
Example

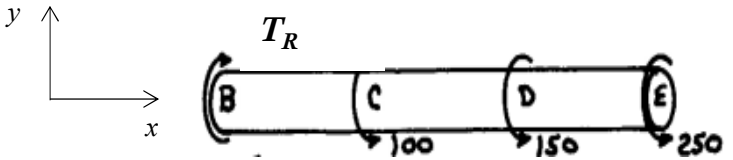
The motor shown below supplies a torque of 500 Nm to shaft BCDE. The torques removed at gears C, D, and E are 100 Nm, 150 Nm, and 250 Nm respectively.

a) Determine the torques transmitted by the transverse cross sections in intervals BC, CD, and DE and sketch a torque diagram

b) Calculate the torsional shear stresses in the shaft assuming it has a uniform cross-section of diameter 50-mm in all sections.

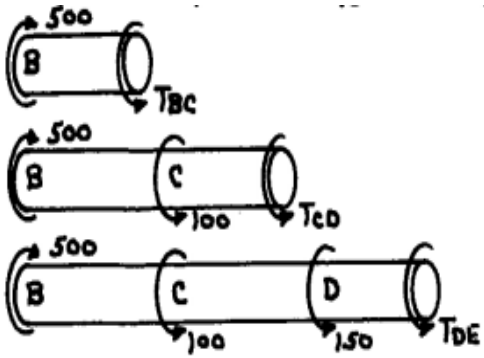
c) In what section does the maximum shear stress occur?





$\circlearrowleft \sum M_x = 0 \; : T_R - 100 - 150 - 250 = 0$

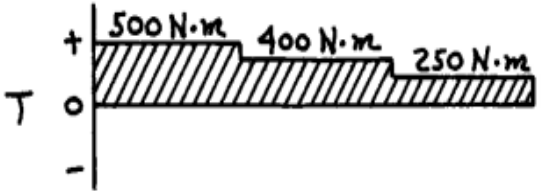
$\therefore \; T_R = 500 \text{ N.m} \circlearrowright$



$\circlearrowleft \sum M = 0 \; : T_{BC} = 500 \text{ N.m} \circlearrowright$

$\circlearrowleft \sum M = 0 \; : T_{CD} = 400 \text{ N.m} \circlearrowright$

$\circlearrowleft \sum M = 0 \; : T_{DE} = 250 \text{ N.m} \circlearrowright$



$J = 6.136 \times 10^{-7} \text{ m}^4$

$\tau_{BC} = 40.7 \text{ MPa}$

$\tau_{CD} = 32.6 \text{ MPa}$

$\tau_{DE} = 20.4 \text{ MPa}$

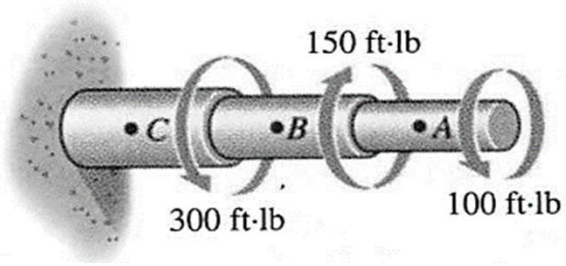
Torsional Loading: Internal Forces

Example

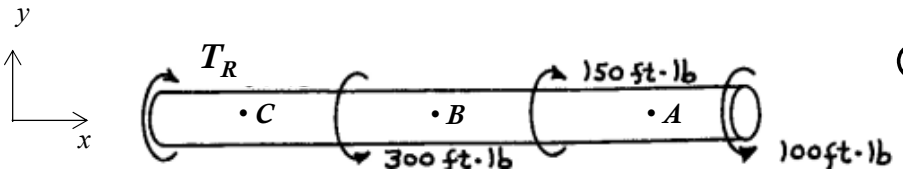
For the steel shaft shown;

a) Determine the torque transmitted by the transverse cross sections at points A, B, and C.

b) Calculate the torsional stress in each section of the bar if the diameter at
C = 15 cm
B = 12 cm
A = 10 cm

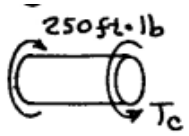


For support reactions on the shaft, recall

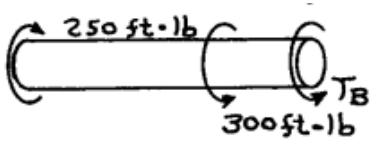


$\circlearrowleft \sum M_x = 0 : T_R - 300 + 150 - 100 = 0$

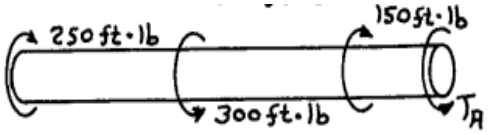
$\therefore \; T_R = 250 \text{ lb.ft} \; \circlearrowright$



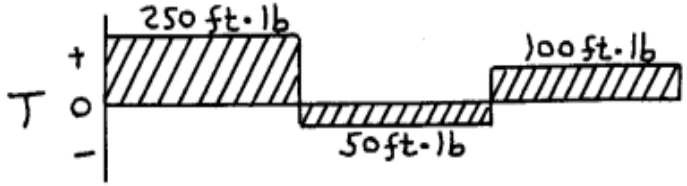
$\circlearrowleft \sum M = 0 : T_C = 250 \text{ ft.lb} \; \circlearrowright$



$\circlearrowleft \sum M = 0 : T_B = 50 \text{ ft.lb} \; \circlearrowright$



$\circlearrowleft \sum M = 0 : T_A = 100 \text{ ft.lb} \; \circlearrowright$



Torsional Loading

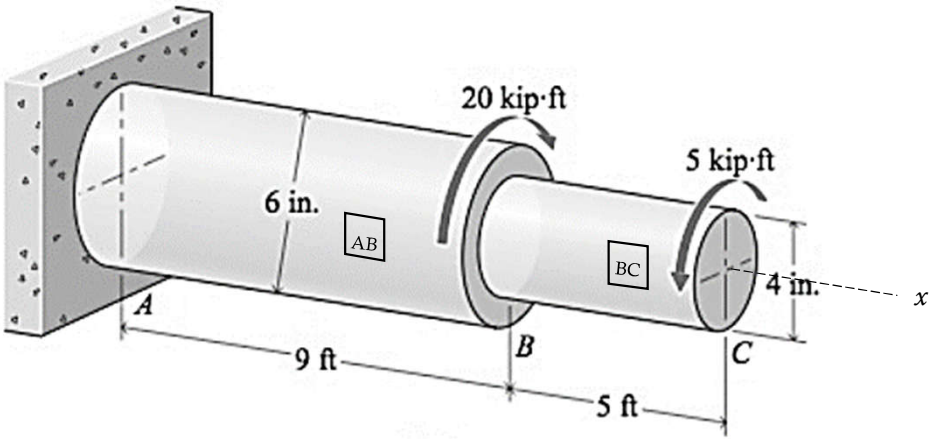
Example

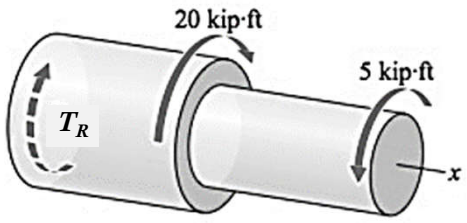
A solid steel shaft attached to a wall and is in equilibrium when subjected to torques as shown. If the Shear Modulus of steel is 12000-ksi, determine;

a) The maximum shear stress in the shaft.

b) Also sketch the stress elements for each section.

c) The total twist in the entire shaft.



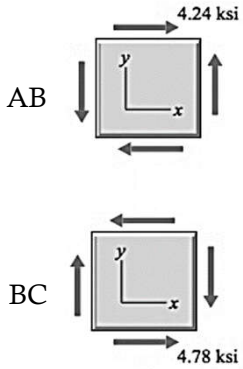


$$+ \quad T_R = 15 \text{ kp.ft} \quad \circlearrowleft$$

$$\tau = \frac{Tr}{J}$$

$$\tau_{AB} = \left(\frac{Tr}{J}\right)_{AB} = 4.244 \text{ ksi}$$

$$\tau_{BC} = \left(\frac{Tr}{J}\right)_{BC} = 4.775 \text{ ksi}$$

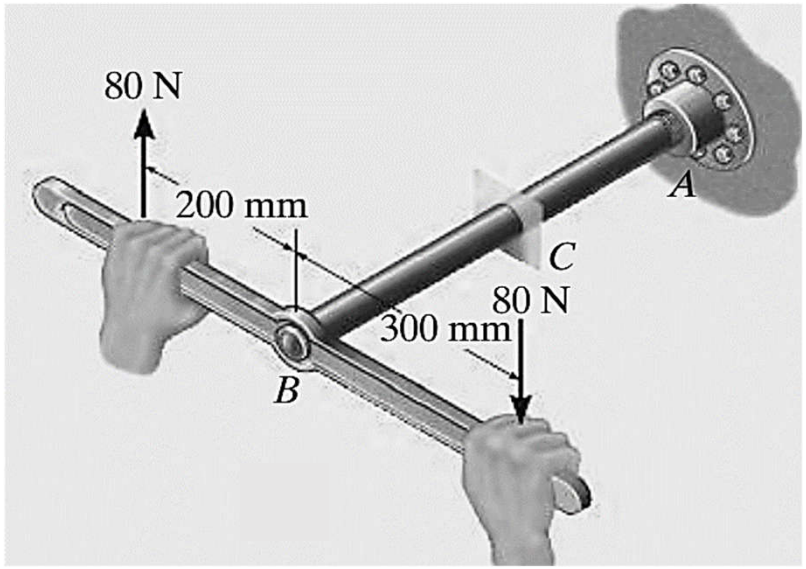


$$\phi_{TOTAL} = \left(\frac{TL}{GJ}\right)_{AB} + \left(\frac{TL}{GJ}\right)_{BC} = 0.000795 \text{ rad} \quad \circlearrowleft$$

Torsional Loading

Example

A pipe has an inner and outer diameters of 80-mm and 100-mm respectively. If its end is tightened against the support at A using a torque wrench at B, determine the shear stress developed in the material at the inner and outer walls at location C of the pipe when the 80-N forces are applied to the wrench



$$\tau = \frac{Tr}{J}$$

$$\circlearrowleft \; T = 80(0.2+0.5) = 40 \text{ N.m} \; \circlearrowright$$

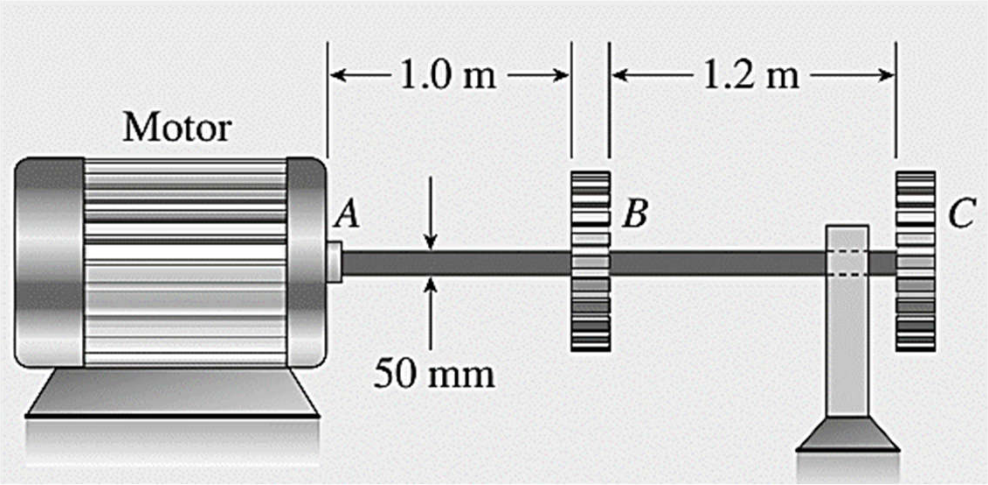
$$J = \frac{\pi}{32}(D_O^{\,4} - D_I^{\,4}) = 5.796\times10^{-6} \text{ m}^4$$

$$\tau_{inner} = 0.345 \text{ MPa} \qquad \tau_{outer} = 0.276 \text{ MPa}$$

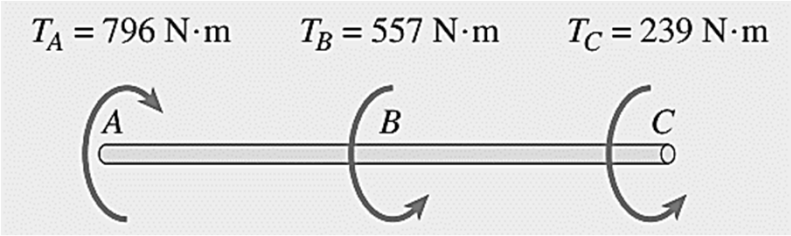
Torsional Loading

Example

A solid steel shaft ABC of 50 mm diameter (Fig. 3-31a) is driven at A by a motor that transmits 50 kW to the shaft at 10 Hz. The gears at B and C drive machinery requiring power equal to 35 kW and 15 kW, respectively. Compute the maximum shear stress τ_{max} in the shaft and the angle of twist ϕ_{AC} between the motor at A and the gear at C. (Use G = 80 GPa.)



$$P = T 2\pi f$$



$$\tau = \frac{Tr}{J}$$

$$J = \frac{\pi}{32}(D^4) = 6.136 \times 10^{-7} \text{ m}^4$$

$$\tau_{AB} = 32.4 \text{ MPa} \qquad \tau_{BC} = 9.7 \text{ MPa}$$

$$\phi = \frac{TL}{GJ}$$

$$\phi_{AC} = \phi_{AB} + \phi_{AB} = 0.0220 \text{ rad}$$

Torsional Loading

Example

A solid steel shaft of diameter 50 mm is to transmit a torque of 75 N.m. If the torsional shears in the shaft must not exceed 120 MPa, determine the maximum speed at which the shaft should rotate.

$\omega = 25.5\ rad/s$

Example

A motor driving a solid circular steel shaft transmits 40 hp to a gear at B shown in the diagram below. The allowable shear stress in the steel is 6000 psi.

a) What is the required diameter *d* of the shaft if it is operated at 500 rpm? *[Ans: D= 1.62 in]*

b) What is the required diameter *d* if it is operated at 3000 rpm? *[Ans: D= 0.89 in]*

